Do Hai Son

Date of birth: August 28^{th}, 1998
Gender: Male
Email: dohaison1998@gmail.com
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Education

2025 – Present **Curtin University** – WA, Australia
PhD student in Electrical and Computer Engineering

2020 – 2023 **VNU University of Engineering and Technology** – Hanoi, Vietnam
Master degree in Communications Engineering

2016 – 2020 **VNU University of Engineering and Technology** – Hanoi, Vietnam
Bachelor degree in Electronics and Communications Engineering

Work experience

Nov 2023 – Present **Information Technology Institute (ITI), Vietnam National University** – Hanoi, Vietnam
Role: Researcher

Aug 2020 – Jul 2023 **Advanced Institute of Engineering and Technology (AVITECH), VNU University of Engineering and Technology** – Hanoi, Vietnam
Role: Researcher

Sep 2022 – Dec 2022 **PRISME Laboratory, École polytechnique de l'université d'Orléans (Polytech Orléans)** – Orléans, France
Role: Internship student

Research experience

Jun 2025 – Present **The Australia - Vietnam Strategic Technologies Centre Project: Strengthening Security in Blockchain and IoT Ecosystems with Data Protection and Intrusion Detection Systems** [Certificate]
Responsibility: Develop a novel framework to detect cyberattacks in blockchain networks.

Sep 2024 – Present **VNU Project: Signal processing for 6G near-field communication and sensing**
Responsibility: A model-driven network for NFC-6G channel estimation.

Mar 2025 – Feb 2026 **Industry Project: A new algorithm for GNSS positioning**
Responsibility: We develop an AI-aided PPP method for faster convergence time.

Jul 2024 – Mar 2025	**Industry Project: Support the development of women-owned businesses in Viet Nam** [**A training session**] Responsibility: We developed a training course on the topic: "Cybersecurity in the digital space for women-owned small and medium-sized enterprises (SMEs)".
May 2022 – Mar 2025	**ASEAN-IVO Project: Agricultural IoT based on Edge computing** [**Work package No. 2**] Responsibility: An agricultural IoT security framework based on authentication, data preservation, and encryption.
Jul 2021 – Oct 2024	**VNU-UET Project: Channel estimation using side information for massive MIMO systems** Responsibility: We considered the antenna structures of massive MIMO and integrated side information (e.g., DoA, DoD) to enhance the performance of estimators.
Jun 2020 – Aug 2023	**NAFOSTED Project: System identification: from blind to informed paradigm** [**Certificate**] Responsibility: We have developed an "InSI" toolbox on MATLAB, which provides a set of tools to analyze, evaluate, and design complicated systems, especially in wireless communications.
Jun 2020 – Nov 2022	**ASEAN-IVO Project: Cyber-attack detection and information security in Industry 4.0** [**Certificate**] Responsibility: We provided tools to enhance cyber-security in Industry 4.0, i.e., blockchain-based smart grid, collaborative learning for cyberattacks detection.
Oct 2019 – Jun 2020	**Graduate Thesis: Direction of Arrival on SDR** Responsibility: We implemented MUSIC algorithm on GNU Radio and SDR devices, i.e., bladeRFx115).
Mar 2019 – Oct 2019	**Self-Project: WiFi Map Indoor Positioning System** Responsibility: We focused on analyzing RSSI data using traditional machine learning methods (i.e., SVM). We then deployed and verified in a robot using Arduino KIT.

Teaching experience

Feb-May/2026	**Lab courses, ELEN2000: Electrical Circuits** (School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University)
Jul-Oct/2025	**Lab courses, ETEN5000: Signals and Systems** (School of Electrical Engineering, Computing and Mathematical Sciences, Curtin University)
Sep-Nov/2023	**Lab courses, AIT 2003: Data Analysis with Python** (VNU University of Engineering and Technology)
Sep-Nov/2023	**Lab courses, INT 1008: Introduction to Programming** (VNU University of Engineering and Technology)

Sep-Dec/2021 **Lab courses, ELT 3243: Principles of Communication** (VNU University of Engineering and Technology)

Jul-Aug/2021 **Lab courses, ELT 2035: Signals and Systems** (VNU University of Engineering and Technology)

Jan-Jun/2021 **Lab courses, ELT 3144: Digital Signal Processing** (VNU University of Engineering and Technology)

Publications

Preprints

1. Tran Viet Khoa, **Do Hai Son**, Mohammad Abu Alsheikh, Yibeltal F. Alem, and Dinh Thai Hoang, "Privacy-Preserving Driver Drowsiness Detection with Spatial Self-Attention and Federated Learning," *IEEE Transactions on Vehicular Technology*, pp. 1–14, Feb. 2026. (under review - round 1) [https://arxiv.org/abs/2508.00287]

Books & Book chapters

1. **Do Hai Son**, Tran Thi Thuy Quynh, Ngo K. Hoang, Nguyen Van Ly, and Nguyen Linh Trung, "Thiết lập nền tảng SDR cho hệ thống OFDM," trong Truyền thông chuyển tiếp hai chiều: Lý thuyết và Thực nghiệm, Nhà xuất bản Đại học Quốc gia Hà Nội, pp. 151-235, 2025.

Journal articles

1. Tran Viet Khoa, **Do Hai Son**, Chi-Hieu Nguyen, Dinh Thai Hoang, Diep N. Nguyen, Tran Thi Thuy Quynh, Trong-Minh Hoang, Nguyen Viet Ha, Eryk Dutkiewicz, Mohammad Abu Alsheikh, and Nguyen Linh Trung, "Collaborative Learning Framework to Detect Attacks in Transactions and Smart Contracts," *IEEE Transactions on Cognitive Communications and Networking*, vol. 12, pp. 4290–4306, Jan. 2026. [IF = 7]

2. Tran Viet Khoa, **Do Hai Son**, Dinh Thai Hoang, Nguyen Linh Trung, Tran Thi Thuy Quynh, Diep N. Nguyen, Nguyen Viet Ha, and Eryk Dutkiewicz, "Collaborative Learning for Cyberattack Detection in Blockchain Networks," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 54, no. 7, pp. 3920-3933, Jul. 2024. [IF = 8.6]

3. **Do Hai Son**, Vu Tung Lam, and Tran Thi Thuy Quynh, "ISDNN: A Deep Neural Network for Channel Estimation in Massive MIMO systems," *Hanoi University of Industry Journal of Science and Technology*, vol. 60, no. 11, pp. 48–54, Nov. 2024.

4. Bui Minh Tuan, Tran Viet Khoa, **Do Hai Son**, Nguyen Linh Trung, Tran Thi Thuy Quynh, Nguyen Ngoc Hoa, Nguyen Viet Ha, Nguyen Dai Tho, and Le Quang Minh, "A New Framework for Cyber Risk Assessment for Industry 4.0 and Recommendations for Vietnam," *REV Journal on Electronics and Communications*, vol. 13, no. 3-4, pp. 28-44, Jul-Dec. 2023.

Conference & Workshop articles

1. **Do Hai Son**, Le Vu Hieu, Tran Viet Khoa, Yibeltal F. Alem, Hoang Trong Minh, Tran Thi Thuy Quynh, Nguyen Viet Ha, and Nguyen Linh Trung, "Vision-Based Learning for Cyberattack Detection in Blockchain Smart Contracts and Transactions," in *International Symposium on Communications and Information Technologies (ISCIT)*, Hanoi, Vietnam, pp. 43-48, Oct. 2025.

2. Ta Giang Thuy Loan, Hoang-Lan Nguyen, Nguyen Thi Ngoc Lan, **Do Hai Son**, Tran Thi Thuy Quynh, Nguyen Linh Trung, Karim Abed-Meraim, and Thanh Trung Le, "Robust Sparse Subspace Tracking from Corrupted Data Observations," in *International Symposium on Communications and Information Technologies (ISCIT)*, Hanoi, Vietnam, pp. 192-197, Oct. 2025.

3. Pham Hai Anh, Tran Trong Duy, **Do Hai Son**, Karim Abed-Meraim, and Nguyen Linh Trung, "FlowEKF: Flow-based Extended Kalman filter," in *Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, Singapore, pp. 1206-1211, Oct. 2025.

4. Vu Tung Lam, **Do Hai Son**, Tran Thi Thuy Quynh, and Le Trung Thanh, "RACNN: Residual Attention Convolutional Neural Network for Near-Field Channel Estimation in 6G Wireless Communications," in *Conference on Information Technology and its Applications (CITA)*, Phnom Penh, Cambodia, pp. 387–399, Jul. 2025.

5. **Do Hai Son**, Bui Duc Manh, Tran Viet Khoa, Nguyen Linh Trung, Dinh Thai Hoang, Hoang Trong Minh, Yibeltal Alem, and Le Quang Minh, "Semi-Supervised Learning for Anomaly Detection in Blockchain-based Supply Chains," in *International Symposium on Communications and Information Technologies (ISCIT)*, Bangkok, Thailand, pp. 140–145, Sept. 2024.

6. **Do Hai Son**, Nguyen Danh Hao, Tran Thi Thuy Quynh, and Le Quang Minh, "W2E (Workout to Earn): A Low Cost DApp based on ERC-20 and ERC-721 standards," in *International Conference on Integrated Circuits, Design, and Verification (ICDV)*, Hanoi, Vietnam, pp. 79–84, Jun. 2024.

7. Tran Viet Khoa, **Do Hai Son**, Dinh Thai Hoang, Nguyen Linh Trung, Tran Thi Thuy Quynh, Diep N. Nguyen, Nguyen Viet Ha, and Eryk Dutkiewicz, "Real-time Cyberattack Detection with Collaborative Learning for Blockchain Networks," in *IEEE Wireless Communications and Networking Conference (WCNC)*, Dubai, UAE, pp. 1-6, Apr. 2024.

8. **Do Hai Son**, Tran Thi Thuy Quynh, and Le Quang Minh, "RANDAO-based RNG: Last Revealer Attacks in Ethereum 2.0 Randomness and a Potential Solution," in *International Workshop on ADVANCEs in ICT Infrastructures and Services (ADVANCE'2024)*, Hanoi, Vietnam, pp. 1-4, Feb. 2024.

9. **Do Hai Son**, Karim Abed-Meraim, Tran Trong Duy, Nguyen Linh Trung, and Tran Thi Thuy Quynh, "On the Semi-Blind Mutually Referenced Equalizers for MIMO Systems," in *Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC)*, Taipei, Taiwan, pp. 278-283, Nov. 2023.

10. **Do Hai Son** and Tran Thi Thuy Quynh, "Impact Analysis of Antenna Array Geometry on Performance of Semi-blind Structured Channel Estimation for massive MIMO-OFDM systems," in *IEEE Statistical Signal Processing Workshop (SSP)*, Hanoi, Vietnam, pp. 314-317, Jul. 2023.

11. **Do Hai Son**, Nguyen Huu Hung, Pham Duy Hung, and Tran Thi Thuy Quynh, "Implementing Channel Estimation in Robotics Networks Using Software-Defined Radio," in *National Conference on Electronics, Communications and Information Technology*, Hanoi, Vietnam, pp. 60-65, Dec. 2023.

12. **Do Hai Son**, Tran Thi Thuy Quynh, Tran Viet Khoa, Dinh Thai Hoang, Nguyen Linh Trung, Nguyen Viet Ha, Dusit Niyato, Diep N. Nguyen, and Eryk Dutkiewicz, "An effective framework of private Ethereum blockchain network for smart grid," in *International Conference on Advanced Technologies for Communications (ATC)*, Ho Chi Minh City, Vietnam, pp. 312-317, Oct. 2021. [**Best Student Paper Award**]

13. **Do Hai Son**, Tran Thi Thuy Quynh, "Synchronize multi-SDRs to implement DoA system," in *National Conference on Electronics, Communications and Information Technology*, Hanoi, Vietnam, pp. 11-16, Dec. 2021.

14. **Do Hai Son**, Tran Duc Manh, and Tran Thi Thuy Quynh, "WiFi Maps Indoor Positioning System," in *National Conference on Electronics, Communications and Information Technology*, Hanoi, Vietnam, pp. 53-57, Dec. 2019.

Honors and scholarships

2025 HDR Scholarship (Tuition & Stipend) of Curtin University for my PhD program.

2024 Certificate of Excellence (VNU Information Technology Institute)
Awarded by the Director for outstanding performance in 2024.

2022 Top 13 AI Tech Matching on the AI4VN 2022: "Collaborative Learning for Cyberattack Detection in Blockchain Networks".

2021 Best Student Paper Award of 2021 International Conference on Advanced Technologies for Communications (ATC).

2020 Excellent Thesis Award (VNU University of Engineering and Technology)
Awarded to the best undergraduate theses from the school.

2019 Certificate & Scholarship for the excellent student awarded (VNU University of Engineering and Technology)
Awarded to top 5 students from the class.

Professional activities

Organizing committee
- **Publication chair**: 18th Asia Pacific Signal and Information Processing Association Annual Summit and Conference (APSIPA ASC), Hanoi, Vietnam, 9-12 Nov. 2026.
- **Technical**: 22nd IEEE Statistical Signal Processing (SSP) workshop, Hanoi, Vietnam, 2-5 Jul. 2023.

TPC member
- IEEE Wireless Communications and Networking Conference (WCNC): 2024, 2025, 2026.

Peer review
- International Conference on Advanced Technologies for Communications (ATC).
- National Conference on Electronics, Communications and Information Technology (REV-ECIT).

Technical skills

Programming languages
Proficient in: Python, MATLAB, JavaScript, Solidity
Familiar with: C/C++, C#, Go

Hardware
Software Define Radio (SDR), IoT Gateway Home/Industrial, Jetson/BeagleBone/Arduino/ESP8266, IoT sensors

Languages

English:
2024 IELTS Academic (6.5/9.0)